

A Data-Driven Overview of Global Renewable Energy Trends

Global Renewable Energy in 2025

How Far Have We Come?

Data: Ember (2026); Energy Institute – Statistical Review of World Energy (2025)
Our World in Data – Hannah Ritchie, Max Roser, Pablo Rosado

75% of Global GHG Emissions Still Come from Fossil Fuels

Since the Industrial Revolution, fossil fuels have dominated the world's energy systems — driving climate change and harming public health.



75%

of global greenhouse gas emissions come from burning fossil fuels for energy



5M+

premature deaths per year caused by fossil fuel air pollution



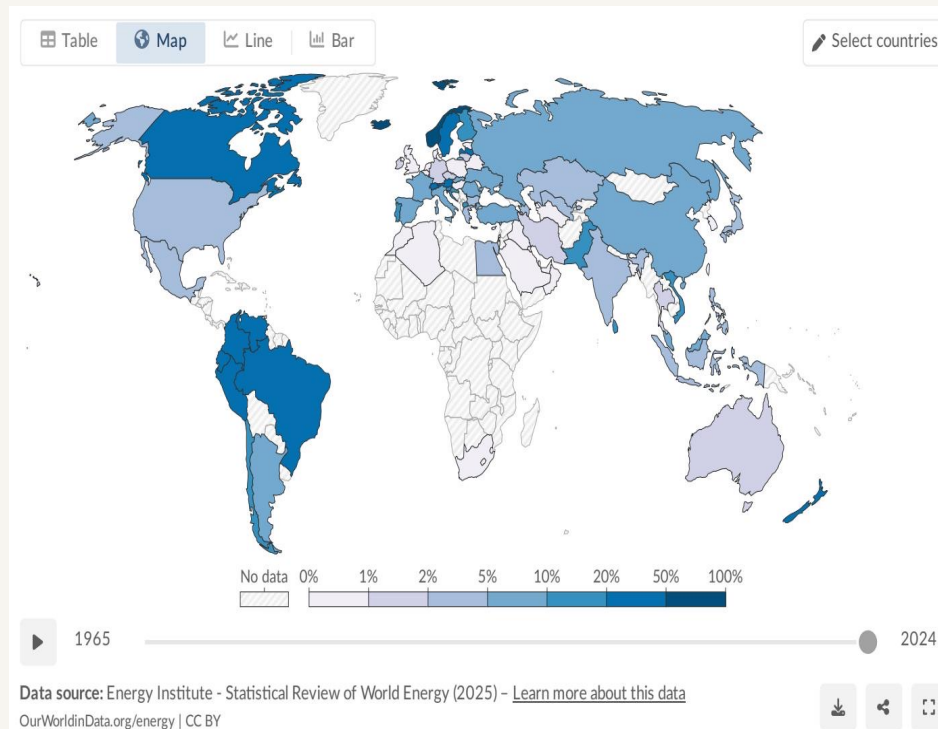
**Since
1850**

fossil fuels have dominated most countries' energy systems, from the Industrial Revolution to today

Renewables Now Supply ~14% of Global Primary Energy

Key Points

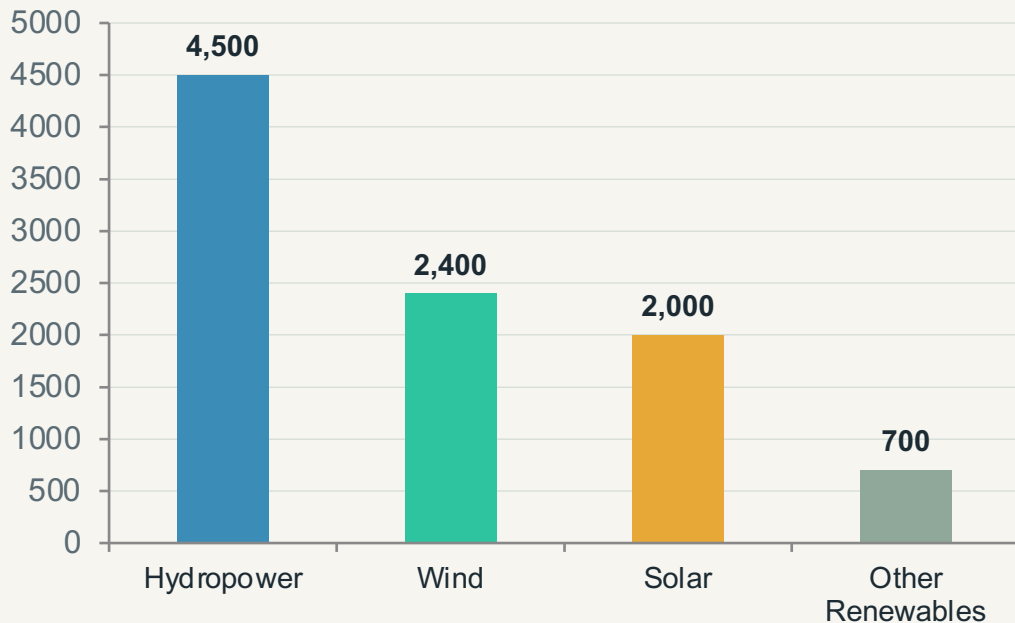
- Approximately one-seventh (~14%) of the world's primary energy comes from renewable technologies (2024, substitution method).
- Primary energy encompasses electricity, transport, and heating combined.
- Transport and heating are harder to decarbonize, keeping the renewable share lower than in electricity alone.
- Countries like Norway, Brazil, and New Zealand lead; many Middle Eastern nations remain below 5%.



Share of primary energy from renewables by country, 2024

Renewable Electricity Generation Is Growing Rapidly

Led by Wind and Solar



4,400 TWh

Wind + Solar combined nearly match hydropower's 4,500 TWh output

Share of Renewable Electricity

Hydropower: ~47%

Wind: ~25%

Solar: ~21%

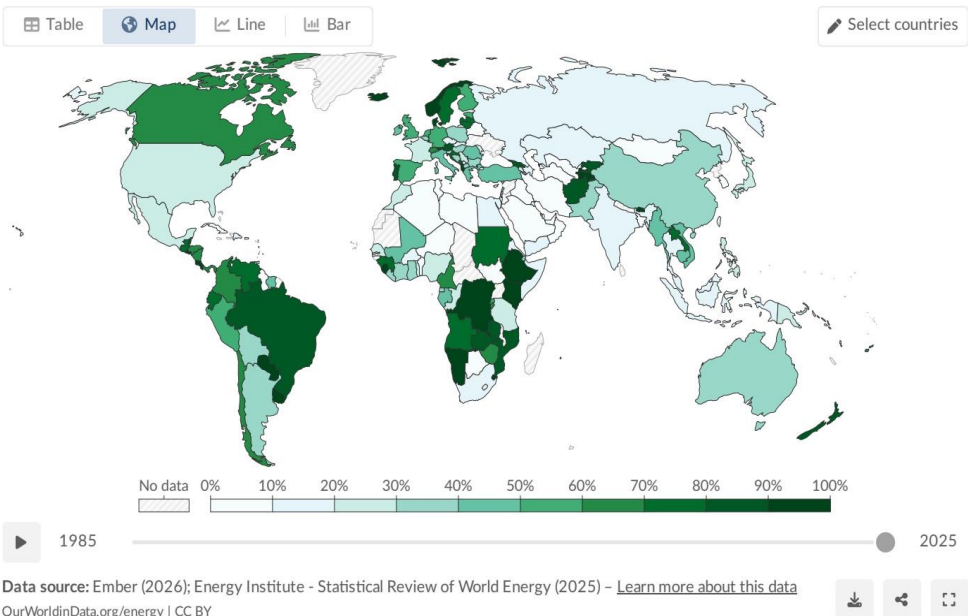
Other: ~7%

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World in Data



Share of electricity from renewables by country, 2025

Region / Country	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World Average	~30%

Several African and Middle Eastern countries have renewable electricity shares below 10%.

Hydropower Remains the Largest Renewable Source

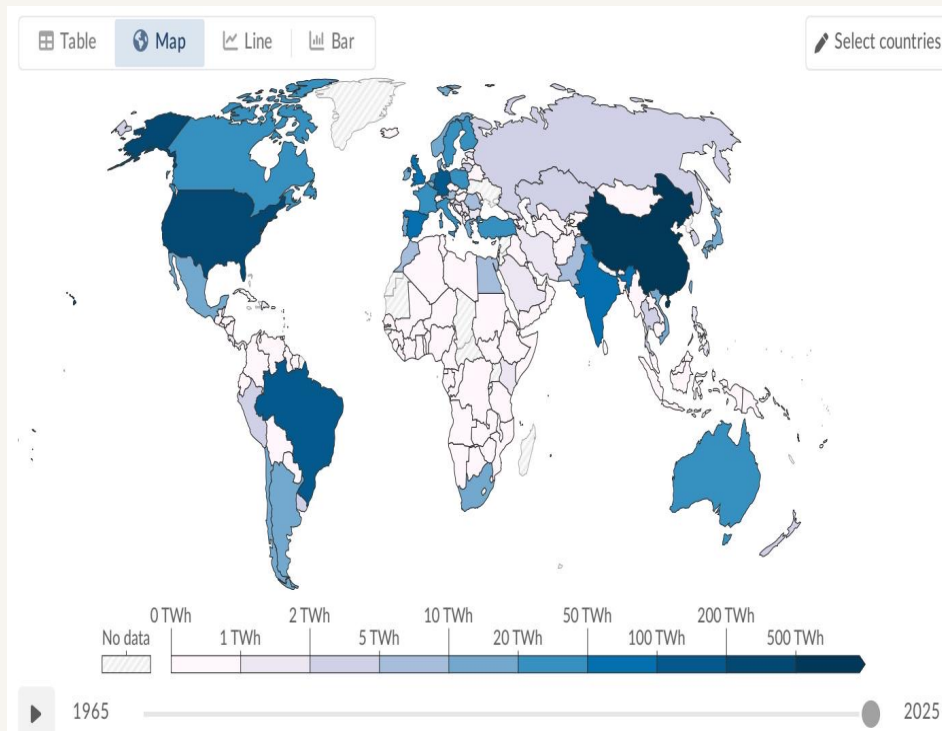
~50

%

of all renewable
electricity generation

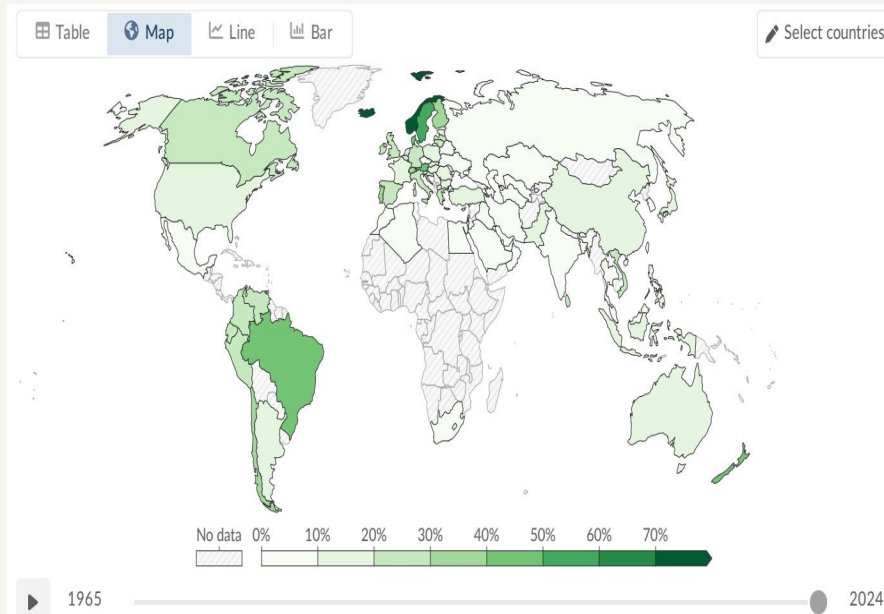
Key Facts

- Generating electricity at scale for over a century
- ~4,500 TWh generated globally (2024)
- Top producers: China, Brazil, Canada
- Growth potential is limited compared to wind and solar



Hydropower generation by country, 2025

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind power generation by country, 2025

Wind Energy

~2,400 TWh generation • ~25% of renewable electricity

Major contributor in Europe, China, and the US (onshore + offshore)

Solar Energy

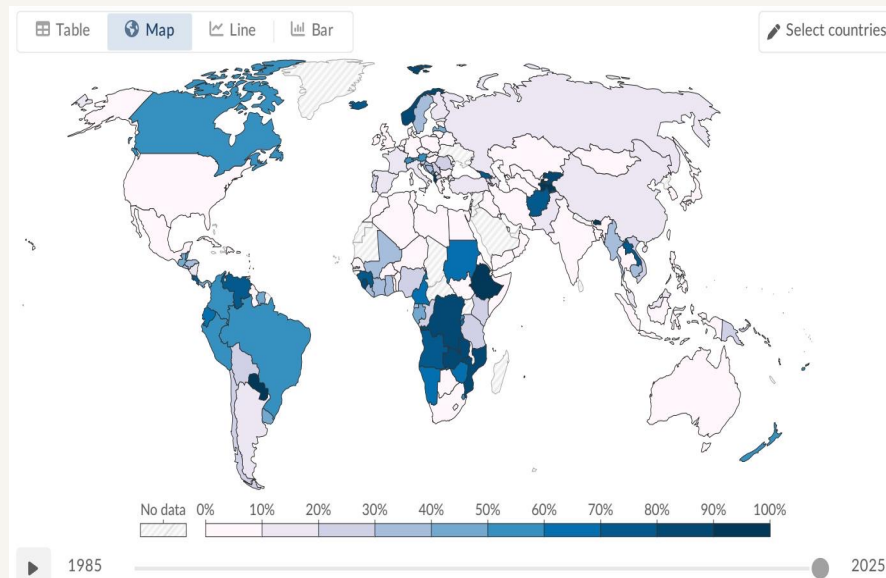
~2,000 TWh generation • ~21% of renewable electricity

Fastest-growing source — dramatic growth since 2010 driven by falling costs and policy support

Wind + Solar combined = ~4,400 TWh
Nearly matching hydropower's output — and growing much faster

Stark Regional Gaps in the Renewable Transition

Sub-Saharan Africa lags while Scandinavia and Latin America lead



Share of electricity from hydropower by country, 2025

Leading Regions (60–98%)

Norway (~98%) — abundant hydropower
Brazil (~85%) — hydro + wind expansion
Canada (~65%) — large hydro resources
Denmark & Germany — heavy wind/solar investment

Lagging Regions (<10%)

Saudi Arabia (<2%) — near-total fossil fuel dependence
South Africa (~10%) — coal-dominated grid
Much of Sub-Saharan Africa — minimal renewable infrastructure
Middle East — almost entirely fossil-fuel dependent for electricity

Key Takeaways

Renewables Are Growing Fast but the Transition Must Accelerate

- 01** ~14% of global primary energy and ~30% of electricity now come from renewables — a milestone, but far from sufficient for climate targets.
- 02** Hydropower remains dominant at ~50% of renewable electricity, but its growth potential is limited compared to wind and solar.
- 03** Wind and solar are the fastest-growing energy technologies in history; combined, they now produce nearly as much electricity as hydropower.
- 04** Stark regional disparities persist: Scandinavia and Latin America exceed 60–90% renewable electricity; much of the Middle East and Sub-Saharan Africa remain below 10%.
- 05** Transport and heating lag far behind electricity in the renewable transition, keeping the overall primary energy share much lower.

The world must accelerate the transition to renewables — **especially in transport, heating, and regions currently left behind.**